

Individual Report

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1 My Contribution

I contributed to the initial ideas for the project and was responsible for most of the secondary encoding, including its Python implementation. Due to my previous industry experience, I also contributed best practices for Python development and GitHub workflows. When parts of the code behaved differently from what we expected, I proposed possible explanations and approaches for resolving these issues.

I was also responsible for running the experiments and documenting their results. For the final paper, I wrote the section describing the Python implementation as well as the results section. I also contributed to the general development and refinement of the project throughout its different stages.

2 Personal Reflection

I particularly liked the freedom we had to choose which aspects of the project to explore. We could either focus on developing more complex Clingo code or extend the Python implementation. At the same time, this freedom was also one of the most challenging aspects of the project. We initially had the idea of investigating track malfunctions, but we were not sure how to develop this idea into a meaningful evaluation.

By looking at previous projects and papers from past students, we came up with the idea of evaluating the running time of the secondary encoding. This turned out to be a successful direction.

Another challenge was finding an effective cadence for working on the project. In the beginning, we were unsure how frequently we should meet. During the lecture period, the weekly meetings provided a good opportunity to stay aligned. After the lecture period ended, we found that more frequent, daily quick stand-ups were useful for keeping the project moving.

If I were to start the project again, I would establish regular weekly meetings from the beginning and increase their frequency towards the end of the semester as the deadline approached.

3 Recommendation to Future Students

Based on my experience, I would recommend that future students spend some time identifying what they genuinely want to explore before committing to a

specific direction. We had two ideas at the beginning and eventually abandoned one in favour of the other. This worked well for us, but such a change should be made consciously rather than simply because there is too little time left in the semester.

I would also recommend having at least one meeting per week where each team member briefly explains what they have completed and what they are currently working on. This makes it easier to identify when someone is stuck and provides an opportunity for the team to help each other. Towards the end of the semester, I would increase the frequency to daily or every second day, particularly when approaching the final implementation and evaluation.